Elementary Analysis

Section 23: Power Series

Exercises 23.1–23.9, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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23 Power series

For a power series $\sum_{n\geq 0} a_n x^n$, write $\beta = \limsup_{n\to\infty} |a_n|^{1/n}$. The radius of convergence is $R = 1/\beta$, with $1/0 = \infty$ and $1/\infty = 0$. The series converges absolutely for $|x| < R$ and diverges for $|x| > R$. *Neither endpoint is decided by the radius calculation*: substitute $x = R$ and $x = -R$ into the original series and test them separately. Finite initial terms do not affect convergence. Thus all series containing $1/n$ start at $n = 1$, and harmless choices of starting index do not change the answers. We use $n^{c/n} \to 1$ for fixed c, which follows by taking logarithms: $\log(n^{c/n}) = c \log n/n \to 0$.

23.1 Exercise 23.1: Eight radii and their endpoint tests

What we must show. Find the exact convergence interval in each case.

(a) The series $\sum n^2 x^n$.

Here $|a_n|^{1/n} = n^{2/n} \to 1$, so $R = 1$. At $x = 1$ the terms are n^2, and at $x = -1$ they are $(-1)^n n^2$. In both cases the terms fail to tend to zero, a necessary condition for series convergence. Thus the exact interval is $\boxed{(-1, 1)}$.

(b) The series $\sum (x/n)^n$.

The coefficient of x^n is $1/n^n$, whose nth root is $1/n \to 0$. Thus $R = \infty$. To see this directly, for fixed real x and sufficiently large n we have $|x|/n \leq 1/2$, and hence $|(x/n)^n| \leq 2^{-n}$. The geometric comparison proves absolute convergence. The exact interval is $\boxed{\mathbb{R}}$.

(c) The series $\sum (2^n/n^2)x^n$.

The coefficient root $2/n^{2/n}$ tends to 2, so $R = 1/2$. At $x = 1/2$ the series becomes $\sum 1/n^2$, which converges. At $x = -1/2$ it becomes $\sum (-1)^n/n^2$, which converges absolutely by the same p-series. The exact interval is $\boxed{[-1/2, 1/2]}$.

(d) The series $\sum (n^3/3^n)x^n$.

The coefficient root $n^{3/n}/3$ tends to $1/3$, giving $R = 3$. At $x = 3$ the terms are n^3; at $x = -3$ they are $(-1)^n n^3$. Neither tends to zero. The exact interval is $\boxed{(-3, 3)}$.

(e) The series $\sum (2^n/n!)x^n$.

For $x \neq 0$ the ratio of successive absolute terms is

$$\frac{2^{n+1}|x|^{n+1}/(n+1)!}{2^n|x|^n/n!} = \frac{2|x|}{n+1} \longrightarrow 0 < 1.$$

The ratio test proves absolute convergence for every such x. At $x = 0$ convergence is immediate. Thus $R = \infty$ and the interval is $\boxed{\mathbb{R}}$.

(f) The series $\sum x^n/((n+1)^2 2^n)$.

Its coefficient root is $1/(2(n+1)^{2/n}) \to 1/2$, so $R = 2$. At $x = 2$ it is $\sum 1/(n+1)^2$; at $x = -2$ it is $\sum (-1)^n/(n+1)^2$. Both converge absolutely. The interval is $\boxed{[-2, 2]}$.

(g) The series $\sum (3^n/(n4^n))x^n$.

The coefficient root $(3/4)n^{-1/n}$ tends to $3/4$, so $R = 4/3$. At $x = 4/3$ we obtain the divergent harmonic series $\sum 1/n$. At $x = -4/3$ we obtain $\sum(-1)^n/n$, which converges by the alternating-series test: $1/n$ decreases to zero. It is conditionally convergent because its absolute-value series is harmonic. The interval is $\boxed{[-4/3, 4/3)}$.

(h) The series $\sum((-1)^n/(n^2 4^n))x^n$.

Taking absolute values before the root removes the alternating sign: $|a_n|^{1/n} = 1/(4n^{2/n}) \to 1/4$. Hence $R = 4$. At $x = 4$ the series is $\sum(-1)^n/n^2$, and at $x = -4$ it is $\sum 1/n^2$, since the two factors $(-1)^n$ multiply to 1. Both converge absolutely, giving $\boxed{[-4, 4]}$.

Conclusion. The radii in order are $1, \infty, 1/2, 3, \infty, 2, 4/3, 4$. The separate endpoint tests determine the displayed intervals.

23.2 Exercise 23.2: Four more series, including missing powers

(a) The series $\sum \sqrt{n}\, x^n$.

The coefficient root is $n^{1/(2n)} \to 1$, giving $R = 1$. At $x = 1$ and $x = -1$, the term magnitudes are $\sqrt{n}$, which do not tend to zero. Thus the exact interval is $\boxed{(-1, 1)}$.

(b) The series $\sum x^n/(n\sqrt{n})$.

The denominator here is $n\sqrt{n} = n^{3/2}$, not an nth root. We have $|a_n|^{1/n} = n^{-3/(2n)} \to 1$, so $R = 1$. At either endpoint the absolute series is $\sum n^{-3/2}$, a convergent p-series because $3/2 > 1$. Thus the exact interval is $\boxed{[-1, 1]}$.

(c) The sparse series $\sum x^{n!}$.

For $|x| < 1$, since $n! \geq n$, $|x|^{n!} \leq |x|^n$. The right side has a convergent geometric sum, so absolute convergence follows. At $x = 1$ every term is 1. At $x = -1$, all terms with $n \geq 2$ are 1 because $n!$ is even. Both endpoint series diverge. For $|x| > 1$, the term magnitudes $|x|^{n!}$ do not approach zero. Therefore the radius is $R = 1$ and the exact interval is $\boxed{(-1, 1)}$.

(d) The series $\sum(3^n/\sqrt{n})x^{2n+1}$.

The exponent is $2n + 1$, so treating $3^n/\sqrt{n}$ as the coefficient of x^n would give the wrong radius. Instead write

$$\sum_{n\geq 1} \frac{3^n}{\sqrt{n}} x^{2n+1} = x \sum_{n\geq 1} \frac{(3x^2)^n}{\sqrt{n}}.$$

For $x \neq 0$, the absolute-term ratio tends to $3x^2$. Thus the series converges absolutely when $3x^2 < 1$ and diverges when $3x^2 > 1$. At $x = 0$ all terms vanish. At $x = 1/\sqrt{3}$, its terms equal $1/(\sqrt{3}\sqrt{n})$; at $x = -1/\sqrt{3}$, they equal $-1/(\sqrt{3}\sqrt{n})$. Each endpoint gives a constant multiple of the divergent p-series $\sum n^{-1/2}$. Thus

$$\boxed{R = 1/\sqrt{3}, \qquad \text{interval } (-1/\sqrt{3}, 1/\sqrt{3}).}$$

Conclusion. Missing powers must be respected. Testing the actual terms, rather than a misindexed coefficient sequence, avoids that error.

23.3 Exercise 23.3: The endpoint test for the sparse series in Example 6

What we must show. Determine the exact interval for $\sum_{n\geq 0} 2^{-n}x^{3n}$.

Recognize a geometric series in a new quantity.

Since $2^{-n}x^{3n} = (x^3/2)^n$, this is geometric with ratio $r = x^3/2$. It converges exactly when $|r| < 1$, equivalently $|x| < 2^{1/3}$. At $x = 2^{1/3}$ the terms are 1; at $x = -2^{1/3}$ they are $(-1)^n$. Neither term sequence tends to zero.

Conclusion. The exact interval is $(-2^{1/3}, 2^{1/3})$. The radius is $2^{1/3}$, not 2, because the powers are x^{3n}.

23.4 Exercise 23.4: Roots and ratios with oscillating coefficients

What we must show. Analyze $a_n = [(4 + 2(-1)^n)/5]^n$.

Separate even and odd indices before calculating.

For even n, $a_n = (6/5)^n$; for odd n, $a_n = (2/5)^n$. The term $a_0 = 1$ does not affect any limiting or convergence question.

(a) Compute the four requested extended limits.

The sequence $a_n^{1/n}$ alternates between $6/5$ and $2/5$, so

$$\limsup a_n^{1/n} = 6/5, \qquad \liminf a_n^{1/n} = 2/5.$$

For even n, the next index is odd and

$$\frac{a_{n+1}}{a_n} = \frac{(2/5)^{n+1}}{(6/5)^n} = \frac{2}{5}\,3^{-n} \longrightarrow 0.$$

For odd n, the next index is even and

$$\frac{a_{n+1}}{a_n} = \frac{(6/5)^{n+1}}{(2/5)^n} = \frac{6}{5}\,3^{n} \longrightarrow +\infty.$$

All coefficients are positive, so absolute values change nothing. Thus the ratio limsup is $+\infty$ and the ratio liminf is 0.

(b) Apply the necessary term test to the two numerical series.

Along even indices, $a_{2k} = (6/5)^{2k} \to +\infty$. Hence a_n does not tend to zero and $\sum a_n$ diverges. Also $(-1)^{2k}a_{2k} = a_{2k}$, so the signed series $\sum(-1)^n a_n$ diverges for the same reason. Alternating signs cannot make a series converge unless its terms tend to zero.

(c) Find the power-series interval.

The root limsup gives $R = 1/(6/5) = 5/6$. At either $x = 5/6$ or $x = -5/6$, the even terms satisfy

$$a_{2k}x^{2k} = (6/5)^{2k}(5/6)^{2k} = 1.$$

Neither endpoint series has terms tending to zero. Therefore

$$\boxed{R = 5/6, \qquad \text{interval } (-5/6, 5/6).}$$

Conclusion. The ordinary ratio limit does not exist. Its extreme subsequences illustrate why the root limsup is the general formula.

23.5 Exercise 23.5: Two coefficient conditions forcing radius at most one

(a) Infinitely many nonzero integer coefficients.

A nonzero integer has absolute value at least 1. There are therefore arbitrarily large indices n with $|a_n|^{1/n} \geq 1$. Consequently $\limsup |a_n|^{1/n} \geq 1$, and the reciprocal-radius formula yields $R \leq 1$. If that limsup is infinite, the radius is zero, which also satisfies the conclusion.

(b) A positive limsup of coefficient magnitudes.

If $\limsup |a_n| > 0$ (including $+\infty$), choose $c > 0$ below this limsup. Infinitely many indices $n_j \to \infty$ satisfy $|a_{n_j}| > c$. At those indices $|a_{n_j}|^{1/n_j} \geq c^{1/n_j}$, and $c^{1/n_j} \to 1$. Thus $\limsup_n |a_n|^{1/n} \geq 1$ and again $R \leq 1$.

Conclusion. The coefficients need not stay large at every index. A single infinite subsequence bounded away from zero already suffices.

23.6 Exercise 23.6: Endpoint asymmetry and nonnegative coefficients

(a) Convergence at the positive endpoint implies convergence at the negative one.

Suppose $a_n \geq 0$ and the finite radius is $R \geq 0$. If $R > 0$ and $\sum a_n R^n$ converges, then

$$\sum |a_n(-R)^n| = \sum a_n R^n < \infty.$$

Hence the series at $-R$ converges absolutely. If $R = 0$, the two endpoints are the same point and the assertion is immediate. The sign hypothesis is doing the work: the positive-endpoint sum is already a sum of absolute values.

(b) Construct the opposite endpoint pattern.

Take

$$\sum_{n\geq 1} \frac{(-1)^{n+1}}{n} x^n.$$

Its coefficient-root limsup is 1, so its radius is 1. At $x = 1$ the alternating harmonic series converges. At $x = -1$ the series is $-\sum 1/n$, which diverges. Thus the exact interval is $\boxed{(-1, 1]}$. These coefficients do not satisfy the nonnegative hypothesis of part (a), so there is no contradiction.

Conclusion. Endpoint behavior depends on signs as well as magnitudes.

23.7 Exercise 23.7: Powers of cosine and pointwise limits

What we must show. Analyze $f_n(x) = (\cos x)^n$ for each fixed real x.

(a) Points that are not multiples of π.

At such a point $-1 < \cos x < 1$. Therefore $|\cos x|^n \to 0$, and $|f_n(x)| \to 0$ implies $f_n(x) \to 0$.

(b) Even multiples of π.

If $x = 2k\pi$ for an integer k, then $\cos x = 1$, so $f_n(x) = 1$ for every n and the limit is 1.

(c) Odd multiples of π.

If $x = (2k+1)\pi$, then $\cos x = -1$, so $f_n(x) = (-1)^n$. The even subsequence is constantly 1 and the odd subsequence is constantly -1. Since they have different limits, the full limit does not exist.

Conclusion. The three cases exhaust the real line. Continuity of every f_n does not even guarantee pointwise convergence at every point.

23.8 Exercise 23.8: Convergence of functions does not imply convergence of derivatives

What we must show. Compare $f_n(x) = \sin(nx)/n$ with its derivative.

(a) Bound the function values independently of x.

For all real x,

$$|f_n(x)| \le 1/n \longrightarrow 0.$$

The squeeze theorem gives $f_n(x) \to 0$ for every x. In fact this estimate proves uniform convergence, because the upper bound is the same for all x.

(b) Differentiate and test a specific point.

The chain rule gives $f_n'(x) = \cos(nx)$. At $x = \pi$, $f_n'(\pi) = (-1)^n$, so the derivatives have no pointwise limit there. Even at $x = 0$ they are always 1, whereas the derivative of the limit function $f \equiv 0$ is 0.

Conclusion. The operation of taking a derivative cannot be passed through a function limit without additional hypotheses. The oscillations get small in height while their slopes do not.

23.9 Exercise 23.9: A vanishing pointwise limit with nonvanishing integrals

What we must show. Analyze $f_n(x) = nx^n$ on $[0, 1]$.

(a) Fix a point strictly below one.

At $x = 0$ all values are zero. If $0 < x < 1$, set $u_n = nx^n > 0$. Its successive ratio satisfies $u_{n+1}/u_n = (1 + 1/n)x \to x < 1$. Choose q with $x < q < 1$. For some N, $u_{n+1} \le qu_n$ for all $n \ge N$. Repeatedly applying this inequality gives $u_{N+j} \le u_N q^j \to 0$. Thus $f_n(x) \to 0$ for every $x \in [0, 1)$. At $x = 1$, however, $f_n(1) = n \to +\infty$, so no finite limit exists there.

(b) Evaluate each integral before taking the limit.

For fixed n, the ordinary power rule for integration yields

$$\int_0^1 f_n(x)\,dx = n\left[\frac{x^{n+1}}{n+1}\right]_0^1 = \frac{n}{n+1} = 1 - \frac{1}{n+1} \longrightarrow 1.$$

Conclusion. The functions approach zero at every point short of the right endpoint, but their integrals approach 1. Mass concentrates closer and closer to 1; pointwise information alone does not justify interchanging limit and integral.